

WELL COMPLETION REPORT (SYNTHETIC)

OIL-KMC-002 - Kumchai-002

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000024
Field / District	Kumchai / Tinsukia
Block	OALP-IV-04
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2013-11-11 / 2014-01-01
Measured / TVD	3240 m / 3192 m
Trajectory	Vertical; inc 4.9 deg; az 320.0 deg
Formation Interval	1520-2381 m
Lithology	Sandstone-shale-coal sequence
Pore Pressure Reference	3075 psi
Fracture Gradient	15.23 ppg
Mud System	PHPA Water-Based Mud
NWIS Well Risk	Moderate (46.4/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	810	Surface
9 5/8 in casing	1879	Intermediate
7 in liner/casing	2851	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2013-12-21 2599 m	Torque Spike	Critical	Torque rose without corresponding pressure anomaly. Mitigation: Use staged parameter reduction to avoid torsional oscillation. Lesson: Avoid simultaneous high WOB and high RPM in correlated interval.
2013-12-08 1840 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Spot graded LCM pill and circulate at reduced rate. Lesson: Track flow balance and pit volume trend before losses become severe.
2013-12-27 2921 m	Cementing Issue	High	Cement returns were below planned percentage. Mitigation: Use staged cementing response when returns deteriorate. Lesson: Volume imbalance is an important cement-placement warning.
2013-11-22 832 m	Kick	Medium	Unexpected background gas rise preceded influx. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2013-11-22 860 m	Stuck Pipe	Medium	Rotary movement deteriorated progressively. Mitigation: Improve hole cleaning, work pipe and back-ream within operating envelope. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2013-12-20 2549 m	Fishing Operation	Medium	Mechanical obstruction prevented normal drilling progression. Mitigation: Confirm fish geometry before selecting retrieval tool. Lesson: Capture retrieval outcome as institutional knowledge.
2013-12-27 2970 m	Kick	Medium	Unexpected background gas rise preceded influx. Mitigation: Re-establish overbalance and monitor pit/flow balance. Lesson: Treat pit gain plus gas increase as an early influx indicator.

Completion Summary

The synthetic well reached 3240 m MD in the Barail interval. The NWIS knowledge record links 11 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.